Rail Car Drivers Enrichment Details

Rail Car Driver (SurfaceIntegrationDriver) Details

Driver Summary

Field	Value
Level	Medium
Justification	The supplied INF gives reliable details about the hardware ID, service, package files, display curve, and registry settings. No official release notes or verified catalog entry for version 120.8.4.0 were found, and no previous, latest, or family-version fallback was confirmed. Category candidates used exact SID/ KIP driverTag matches where present, then were rejected when their described accessory test scopes did not match the driver's actual functions.

Driver Details

Field	Value
Driver Name	SurfaceIntegrationDriver
Driver Class Name	System
Driver Class Guid	{4D36E97D-E325-11CE-BFC1-08002BE10318}
Driver SC Item Number	5531566
Driver Model	
Driver Provider	%ManufacturerName%
Driver Release Date	05/06/2026
Driver Release Version	120.8.4.0

Extracted Driver Details

Driver Package Name: Surface Integration Driver
Version: 120.8.4.0

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Release Date: 2026-05-06

Driver Component Details

- Surface Integration Driver kernel service (SurfaceIntegrationDriver.sys), demand-start
- Surface ACPI device ACPI\MSHW0530
- Surface power-policy provisioning packages for default, Windows 10, and Windows 11 configurations
- Windows display energy-estimation backlight power curve
- KIP support and Windows/BitLocker registry settings are defined in the INF, but the supplied install sections do not reference the section containing those settings

Feature Keywords

- Surface platform integration
- ACPI\MSHW0530
- SurfaceIntegrationDriver.sys
- Windows 10 build 19041 or later, 64-bit
- Surface power-policy provisioning packages
- KIP power-saving support setting
- 14-point brightness-to-backlight-power curve
- Windows display energy estimation
- battery discharge-time estimate setting
- screen refresh after long power-button press setting
- BitLocker encryption/decryption request limits

Associated Tags

- SID
- KIP

Key Components

- SurfaceIntegrationDriver.sys, installed as a demand-start kernel driver
- ACPI device match ACPI\MSHW0530
- Surface.Power.Policy.Defaults.ppkg
- Surface.Power.Policy.WIN10.Defaults.ppkg
- Surface.Power.Policy.WIN11-0.Defaults.ppkg
- 14-point Windows display backlight power curve, with brightness points from 0% to 100%
- Separate INF registry section defining PlatformKipSupported, battery-discharge estimate, power-button screen-refresh, and BitLocker request-limit settings; the supplied install-section references do not show that section being applied

Key Functionalities

- Installs the Surface Integration Driver system service for the INF-declared ACPI\MSHW0530 device.
- Copies default and Windows-version-specific Surface power-policy provisioning packages and associates their paths with device properties.
- Registers a 14-point brightness-to-backlight-power curve for Windows display energy estimation.
- Defines PlatformKipSupported=1, with an INF comment stating that it enables KIP power-saving

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features; the setting's installation is not confirmed by the shown AddReg references.
• Defines settings to hide the estimated remaining battery discharge time and enable screen refresh after a long power-button press; their installation is not confirmed by the shown AddReg references.
• Defines BitLocker encrypt, decrypt, and per-I/O crypto request limits of 0x40; the INF comment says the encryption request limit is intended to improve SSD performance under BitLocker, but application of these settings is not confirmed.

Impact Details

• Directly affects installation of the Surface Integration kernel service and default/OS-specific power-policy package files for eligible ACPI\MSHW0530 devices.
• May affect Windows display energy estimates through the registered brightness-to-backlight-power curve.
• The INF defines KIP power-saving support, but the provided install-section links do not establish that this setting is applied.
• The INF defines battery discharge estimate and long-press screen-refresh settings, but the provided install-section links do not establish that these settings are applied.
• The INF defines BitLocker request limits and comments on intended SSD performance under BitLocker, but the provided install-section links do not establish that the settings are applied.
• No supported Surface SKU names or version-specific fixes are provided by the INF or verified release information.

Major Components Functionalities

This enrichment uses the exact supplied version 120.8.4.0 and its INF date of 2026-05-06; no exact, previous, latest, or family-version release notes were verified. INFDescriptionData was semantically analyzed and identifies a 64-bit Windows System-class driver for ACPI\MSHW0530, installed as a demand-start kernel service. The package copies default and Windows 10/Windows 11 power-policy provisioning packages and registers a 14-point display backlight curve for energy estimation. Other INF sections define KIP, battery-estimate, screen-refresh, and BitLocker request settings, but the displayed install-section references do not link to that section, so they are not claimed as applied behavior. No official version-specific fixes or changes can be stated from the available sources.

Description Text

The supplied INF identifies Surface Integration Driver version 120.8.4.0, dated 2026-05-06, for the System class and ACPI\MSHW0530 on 64-bit Windows beginning at build 19041. INFDescriptionData was semantically analyzed: the package installs a demand-start kernel service, three named Surface power-policy packages, and a 14-point display backlight power curve under Windows energy estimation. It also contains settings for KIP support, battery estimate visibility, long-press screen refresh, and BitLocker request limits, but the INF's shown install sections do not reference the section containing those settings, so their application cannot be confirmed. The supplied Intel portals concern Intel Management Engine/chipset software and do not identify this Surface package. The Microsoft Update Catalog searches for version 120.8.4.0 and the package name did not expose a verifiable matching Surface entry or release notes; no previous, latest, or family-version release notes were verified or used.

Sources Details

• https://www.intel.com/content/www/us/en/download/682431/intel-management-engine-drivers-for-

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windows-10-and-windows-11.html
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